A new data structure for monomial ideals with applications in various domains of algebra

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Joint work with Pierre Lairez & Rafael Mohr
MATHEXP, Inria Saclay, France

PhD students' seminar
24th February 2026

Plan of the talk

Motivation: Signature Gröbner basis computations

Maximal sharing

Application 1: Hilbert series

Application 2: In commutative algebra

Gröbner bases?

Take a polynomial system:

$$\begin{cases} f_1(x_1, \ldots, x_n) & = & 0 \\ \vdots & & \\ f_s(x_1, \ldots, x_n) & = & 0 \end{cases}$$

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- Applications:
  - polynomial ideal membership,
  - elimination (solve / project polynomial systems),
  - Hilbert series.

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- Observations:
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  - More symbolic preprocessing (membership tests)

Processing of signatures

$\mathbf{eco}^{\mathbf{14}}$: 14 polynomials in 14 variables

(F5 algorithm)

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Number of membership tests in a monomial ideal:

117,861,802

Membership test

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If n number of variables and N number of generators of I then:

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Complexity with our data structure

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$\boxed{\bullet \; 0 \; 2 \; 3}$

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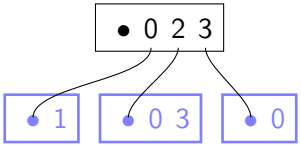

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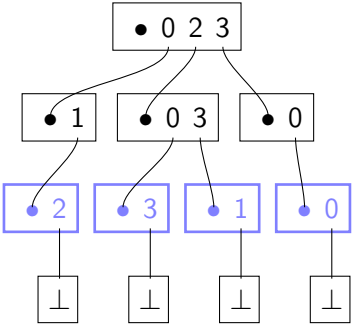

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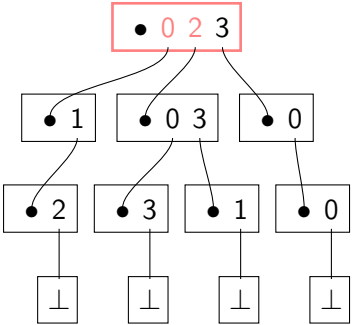

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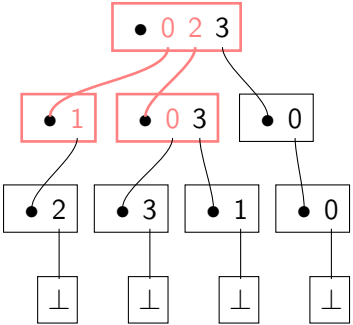

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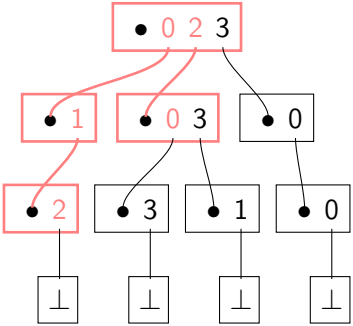

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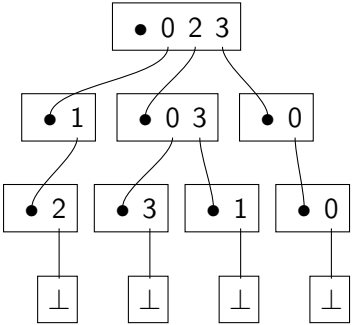
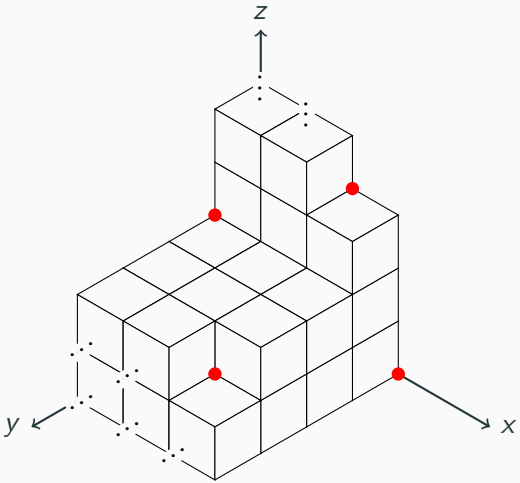


- Roune and Stillman in 2012: kd-trees.

Geometric representation: staircase of $I = \langle yz^2, x^2z^3, x^2y^3z, x^3 \rangle$



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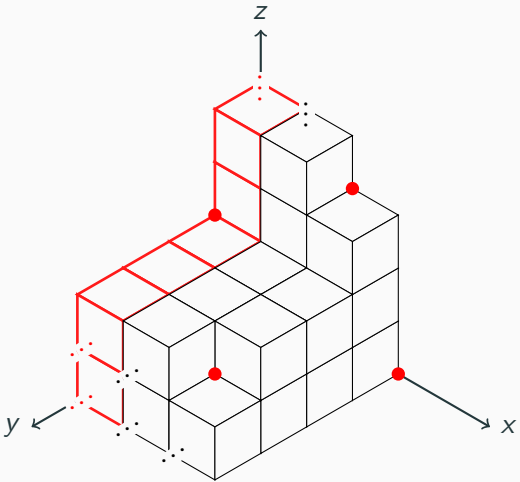

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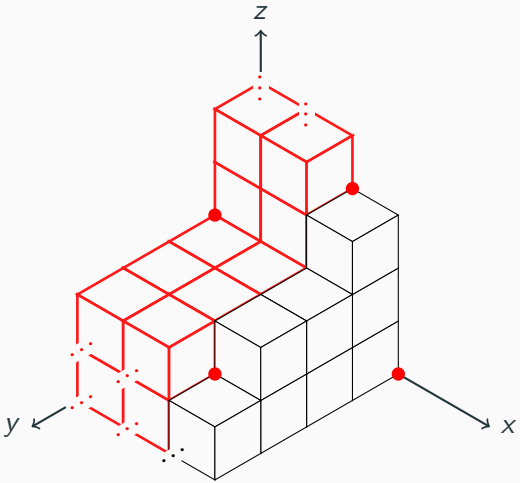

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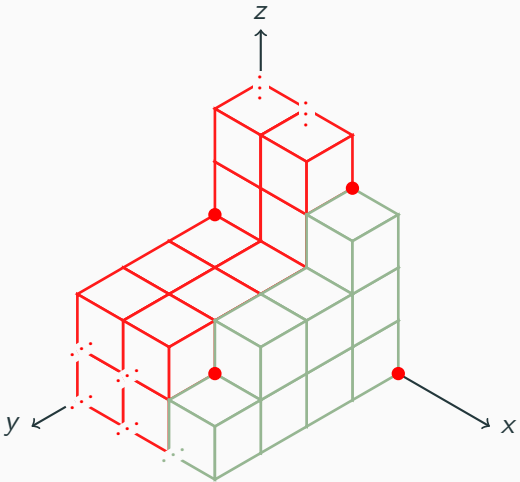

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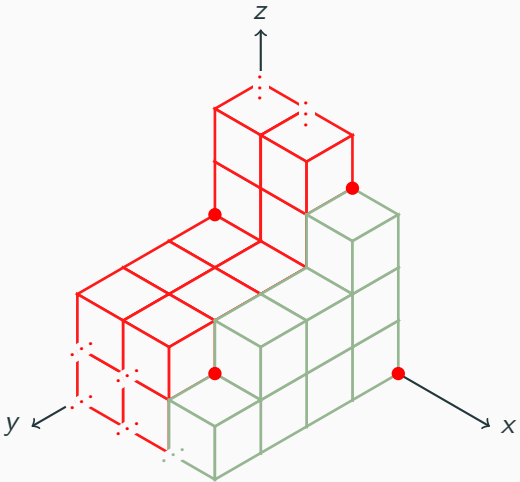

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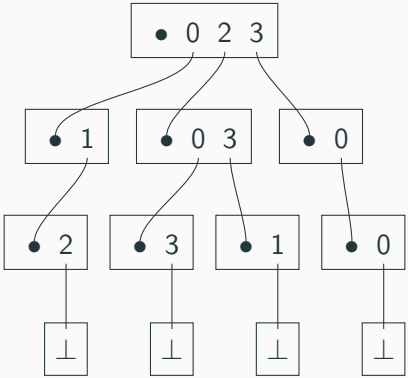

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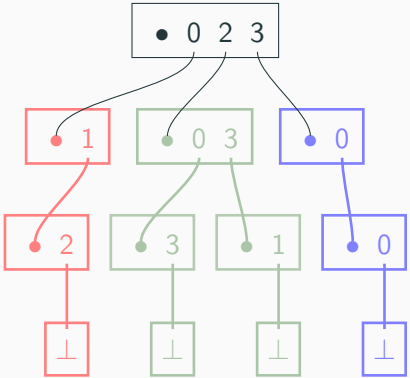

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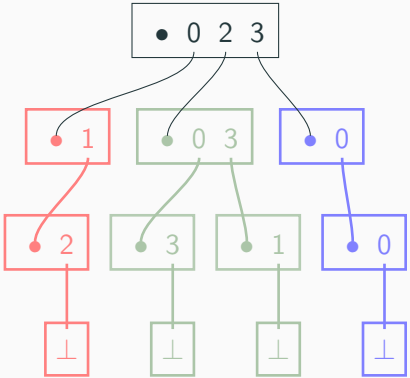

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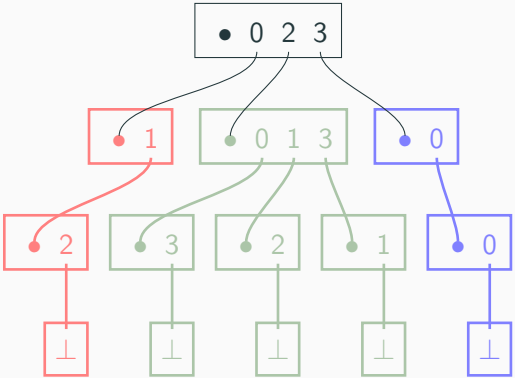

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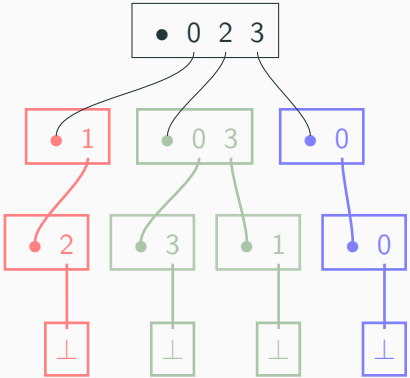

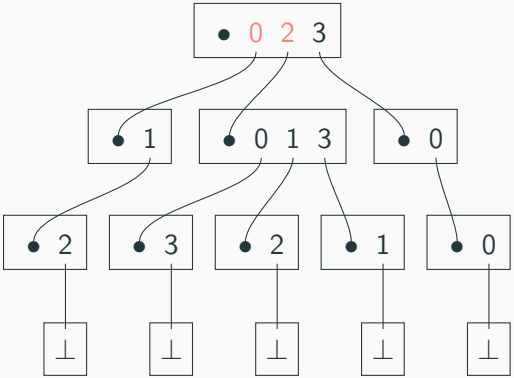

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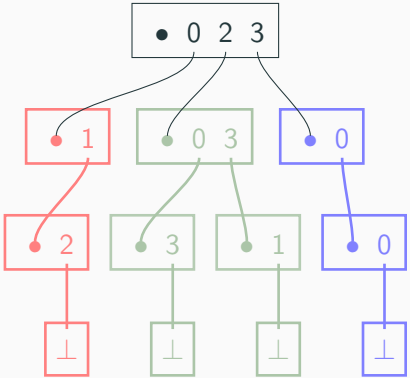

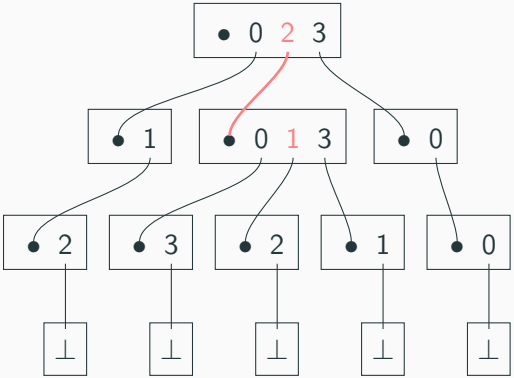

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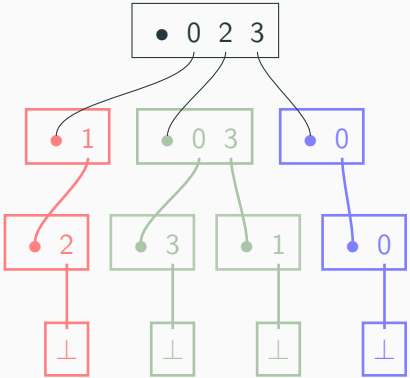

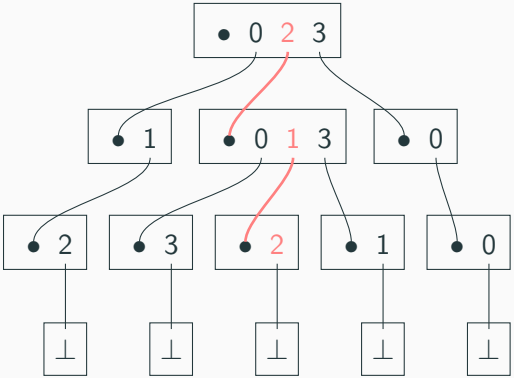

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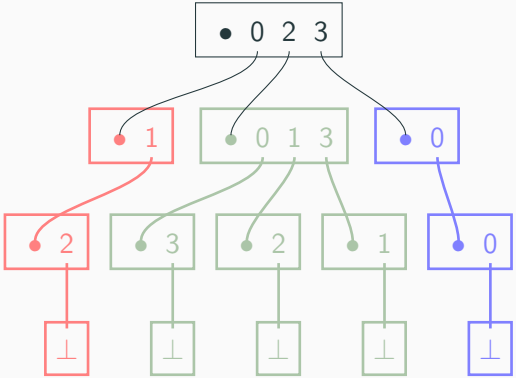

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Ideal n-trees

A tree $t \in \mathcal{T}_n$ is ideal whether $n = 0$ or $n > 0$ and

- $\forall e \in \mathrm{dom}(t)$,

$$t(e) \text{ is ideal}$$

- $\forall e, f \in \mathrm{dom}(t)$,

$$e < f \implies \mathbb{I}(t(e)) \subsetneq \mathbb{I}(t(f))$$

Membership in an ideal n-tree

def `contains`(t, m)

Input: An ideal n-tree t representing I and $m = {x_1}^{m_1} \dots {x_n}^{m_n}$

Output: True if $m \in I$ and false otherwise

```
if n = 0
    return True
s ← t(f) the maximal f ∈ dom(t) s.t. f ≤ m_1
if s = ∅
    return False
return contains(s, x_2^{m_2} ... x_n^{m_n})
```

Existence and uniqueness of ideal n-trees

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There is a one-to-one correspondance between ideal n-trees and non empty monomial ideals.

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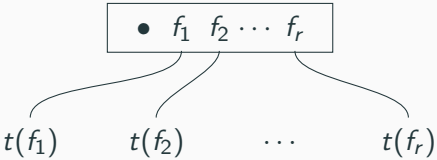

Insertion in an ideal n-tree

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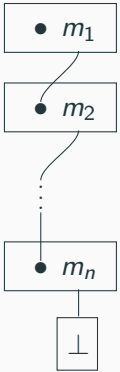

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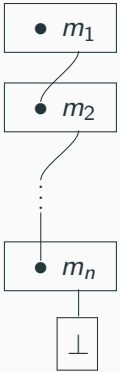


- Let $\texttt{singleton}(m)$ be the ideal n-tree representing $\langle m \rangle$:
- Let $\texttt{insert}(t, m)$ be the ideal n-tree representing $\mathbb{I}(t) + \langle m \rangle$.

Insertion in an ideal n-tree

Two important observations if $m = {x_1}^{m_1}\, m'$:

- For all $e < m_1$,

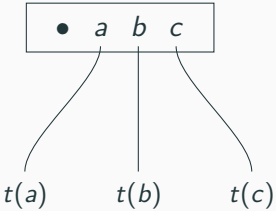
$$(\mathbb{I}(t) + \langle m \rangle)/e \quad = \quad \mathbb{I}(t)/e$$

- For all $e \geq m_1$,

$$(\mathbb{I}(t) + \langle m \rangle)/e \quad = \quad \mathbb{I}(t)/e + \langle m' \rangle$$

Insertion of $m = {x_1}^{m_1}\, m'$:

before insertion:



Insertion of $m = x_1{}^{m_1}\, m'$:

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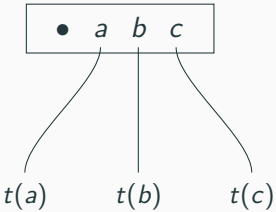
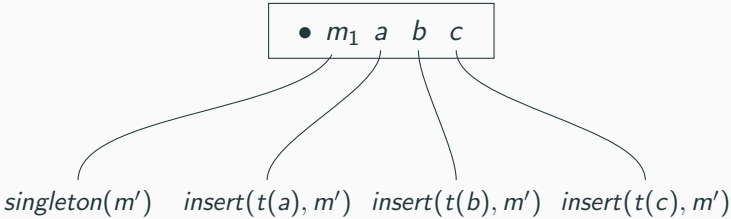


case $m_1 < a$:



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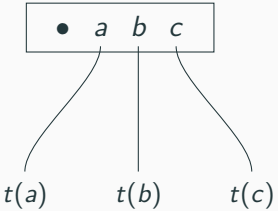

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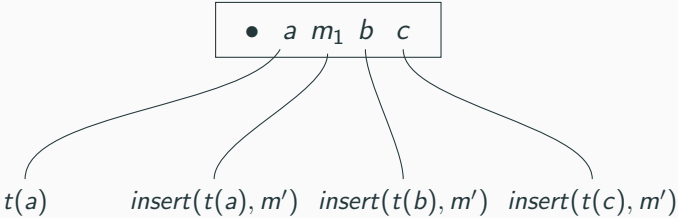

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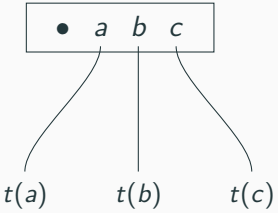
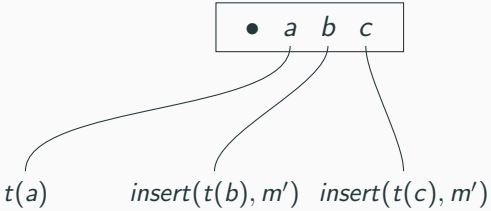
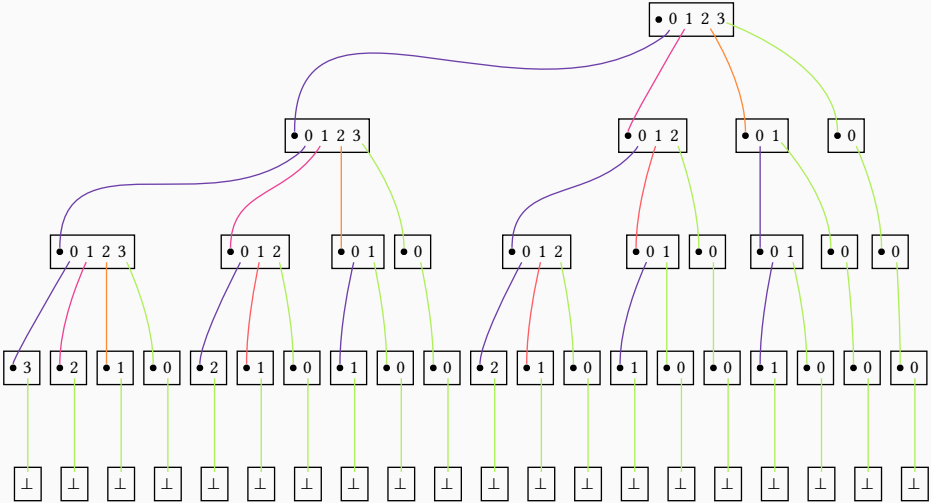


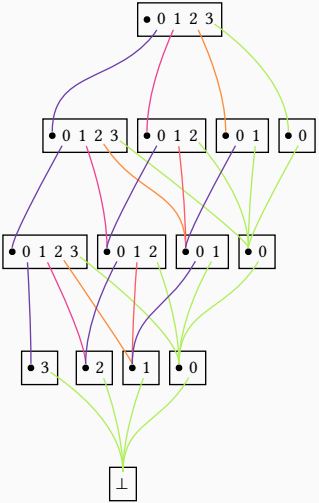
case $m_1 = b$:



Problem of width for ideal n-trees



Solved problem of width with maximal sharing



One example (from Gröbner basis computations)

Number of gen: 5588 Number of nodes: 222 List size: 61 468

Maximal Sharing

Invariant: Two trees are *structurally equal* iff they are *physically equal*.

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Result: Manipulation of ideal n-trees much faster and with less memory.

And what about the size of MDDs?

Two types of monomial ideals appear in GB computations:

- **Weakly revlex:** d maximal degree of a generator of I then MDD size:

$$O(n\,d)$$

- **Borel-fixed:** If N the number of generators of I then MDD size:

$$O(n\,N)$$

Signature Gröbner basis computations

	AlgebraicSolving.jl	*AlgebraicSolving.jl* with mdd					
name	time	time	test	mdd	size	nbr test	×
katsura^{11}	4s	2s	0.04s	0.06s	81	387705	2.0
katsura^{12}	27s	14s	0.17s	0.27s	103	1503655	1.9
katsura^{13}	3min30	1min45	0.71s	1.14s	125	5863729	1.9
cyclic^{7}	1s	0.8s	0.02s	0.02s	43	239572	1.6
cyclic^{8}	1min	29s	0.5s	0.36s	80	4145249	2.1
eco^{14}	3h30	1h06	6s	28s	173	117861802	3.1
noon^{7}	10s	3s	0.08s	0.02s	71	985000	3.1
noon^{8}	3min	44s	0.63s	0.1s	101	7400000	3.5
noon^{9}	78s	83s	0.58s	0.1s	71	7200000	0.9

AlgebraicSolving.jl: Christian Eder (Kaiserslautern), Mohab Safey El Din (Sorbonne Université) and Rafael Mohr (Leuven).

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Application 1: Hilbert series

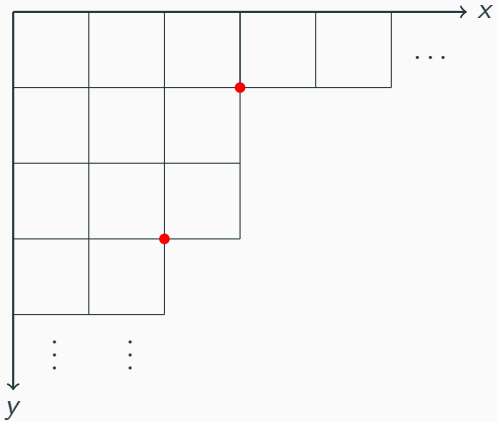
Hilbert series

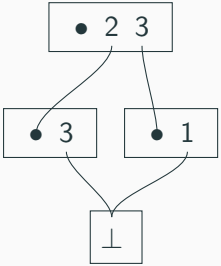
Let $d \in \mathbb{N}$, A_d the set of monomials of degree d outside of I.

The Hilbert series of I is:

$$H_I(t) := \sum_{d \geq 0} |A_d| \, t^d = \sum_{m \notin I} t^{\deg(m)}$$

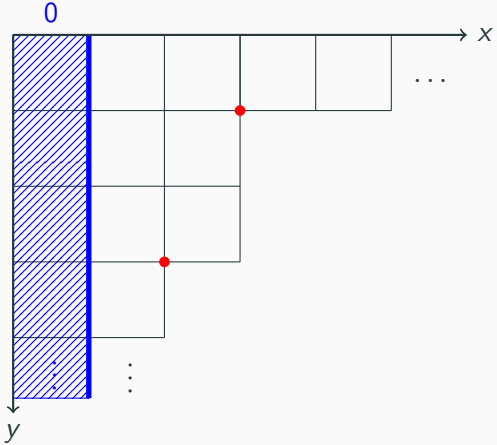
Application 1: Hilbert series of $\langle x^2y^3, x^3y \rangle$

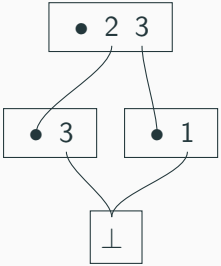




$$H_I(t) =$$

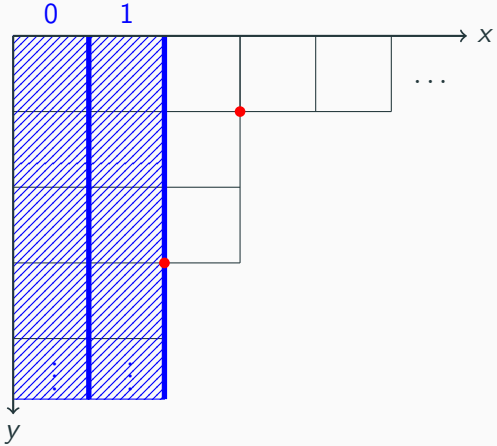
Application 1: Hilbert series of $\langle x^2y^3, x^3y \rangle$

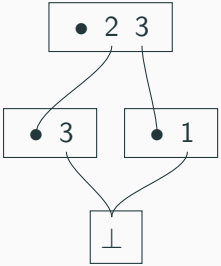




$$H_I(t) = t^0\, H_\varnothing(t)$$

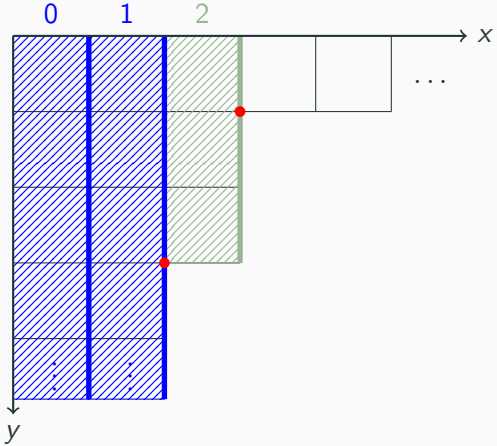
Application 1: Hilbert series of $\langle x^2y^3, x^3y \rangle$

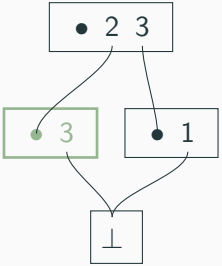




$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t)$$

Application 1: Hilbert series of $\langle x^2y^3, x^3y\rangle$





$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t) + t^2\, H_{\langle y^3\rangle}(t)$$

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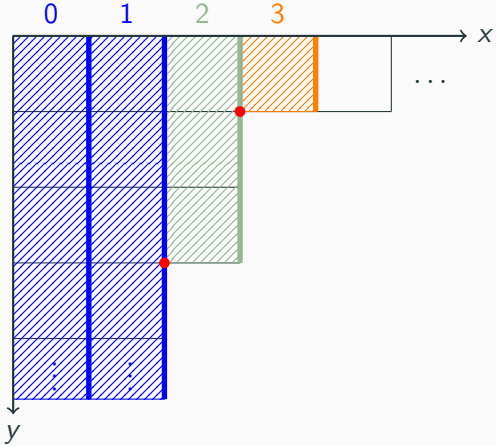

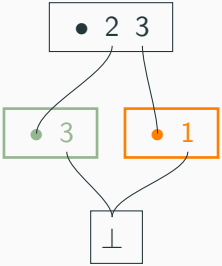

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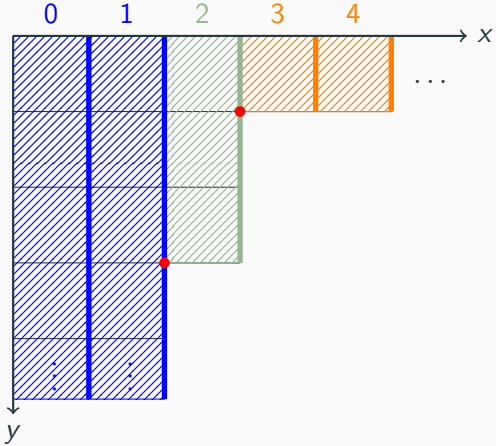

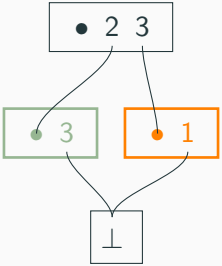

$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t) + t^2\, H_{\langle y^3\rangle}(t) + t^3\, H_{\langle y\rangle}(t) + t^4\, H_{\langle y\rangle}(t)$$

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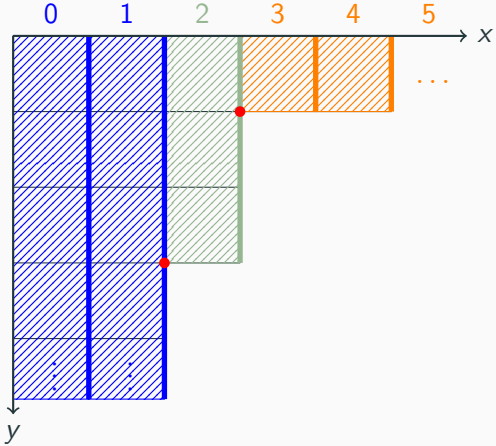

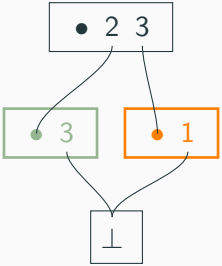

$$H_I(t) = t^0\, H_\varnothing(t) + t^1\, H_\varnothing(t) + t^2\, H_{\langle y^3\rangle}(t) + t^3\, H_{\langle y\rangle}(t) + \sum_{k\geq 4} t^k\, H_{\langle y\rangle}(t)$$

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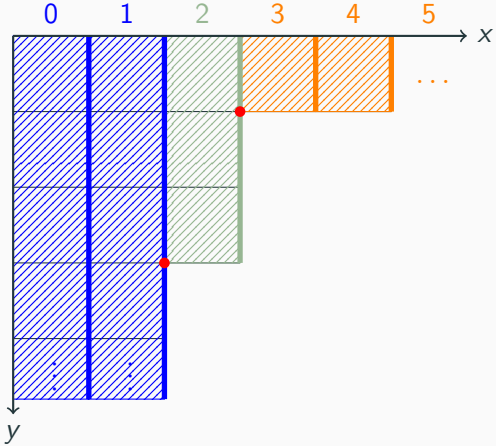
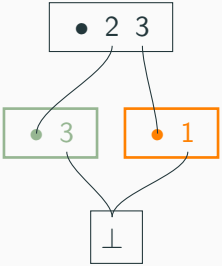
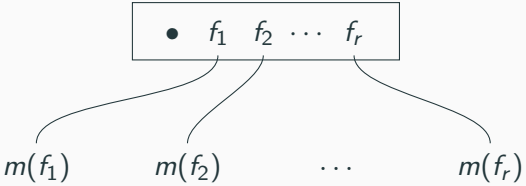




$$H_I(t) = \frac{1-t^2}{(1-t)^2} + \frac{t^2-t^3}{1-t} H_{m(2)} + \frac{t^3}{1-t} H_{m(3)}$$

Application 1: Hilbert series



$$H_I(t) = \frac{1 - t^{f_1}}{(1-t)^n} + \sum_{k=1}^{r-1} \frac{t^{f_k} - t^{f_{k+1}}}{1-t} H_{m(f_k)} + \frac{t^{f_r}}{1-t} H_{m(f_r)}$$

Application 1: Hilbert series

			frobby	monid.rs		
name	gen	var	time	time	size	×
katsura[12]	1050	12	400ms	0.6ms	103	700
katsura[13]	2091	13	1s	1ms	125	1k
eco[14]	2852	14	4s	2ms	173	2k
V10g400	400	10	10s	1s	536329	10
V10g800	800	10	35s	3s	1106025	10
V10g1000	1000	10	50s	4s	1461567	10
frobn13d11	170835	13	18h	11min	34763212	100
J51	3036	89	>1d	14ms	1605	>6m
J60	3432	89	>1d	17ms	1798	>5m
K5	13313	32	>1d	10s	1098244	>8k
tcyc5d25p	3000	125	>1d	16ms	1181	>5m
tcyc5d30p	4350	150	>1d	25ms	1431	>3m
V20g100	100	20	12ms	5ms	6374	2

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V10g1000	1000	10	50s	4s	1461567	10
frobn13d11	170835	13	18h	11min	34763212	100
J51	3036	89	>1d	14ms	1605	>6m
J60	3432	89	>1d	17ms	1798	>5m
K5	13313	32	>1d	10s	1098244	>8k
tcyc5d25p	3000	125	>1d	16ms	1181	>5m
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Application 2: Duality

Idea: Write I a monomial ideal as intersection of irreducible ideals.

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Irreducible monomial ideals

All of the form:

$$\mathfrak{m}_{\underline{a}} = \left\langle {x_1}^{a_1}, \ldots, {x_n}^{a_n} \right\rangle$$

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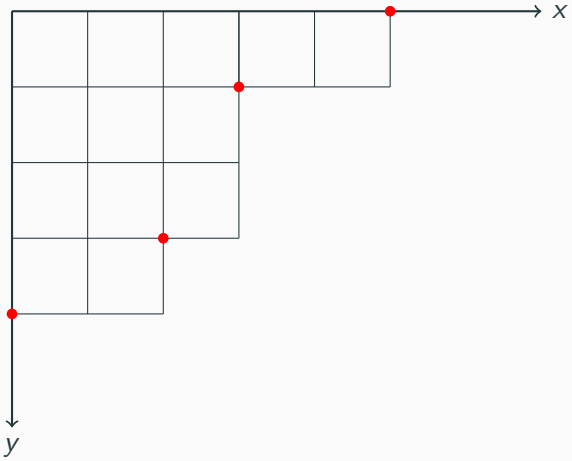
Irreducible components

There exists unique $(\underline{a}_1, \ldots, \underline{a}_r)$ s.t.

$$I = \bigcap_{j=1}^{r} \mathfrak{m}_{\underline{a}_j}$$

Application 2: Duality



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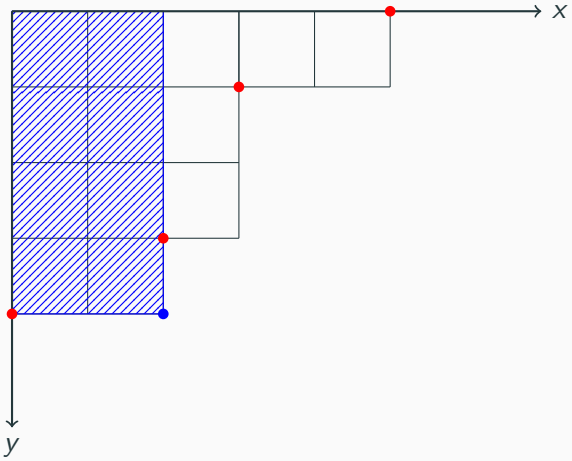

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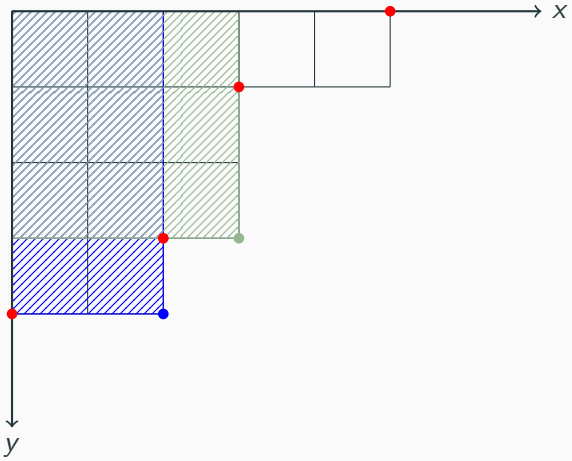

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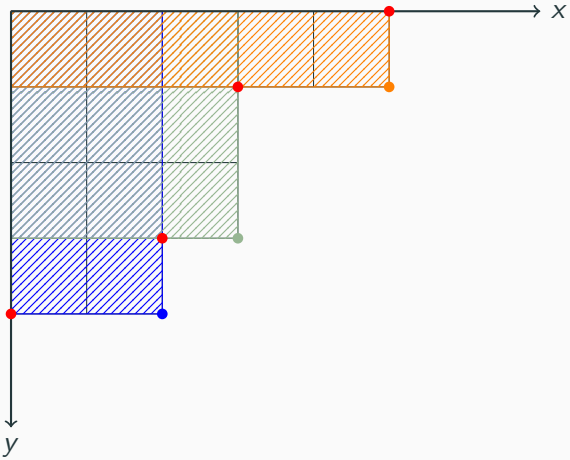

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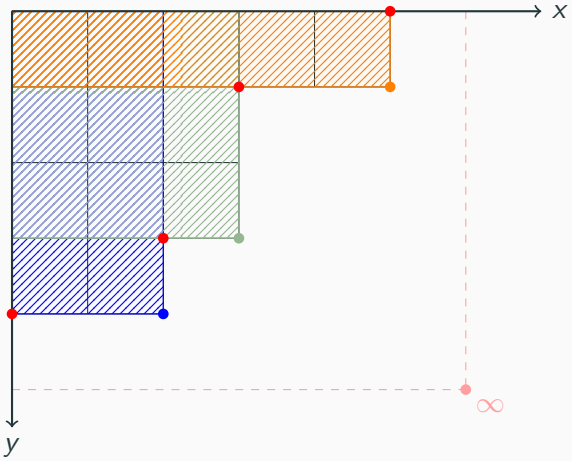

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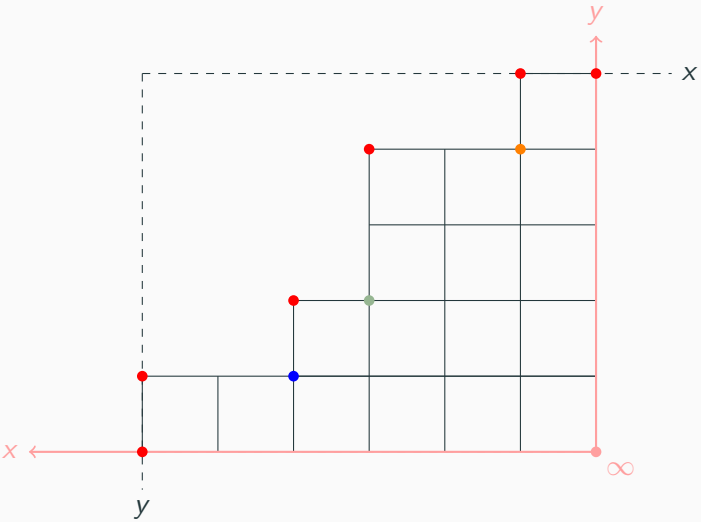
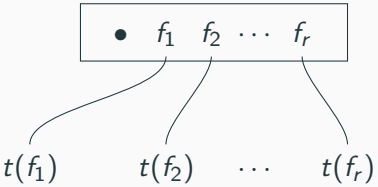


Application 2: Duality

Ideal n-tree I:



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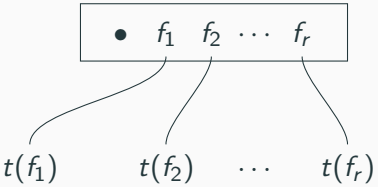
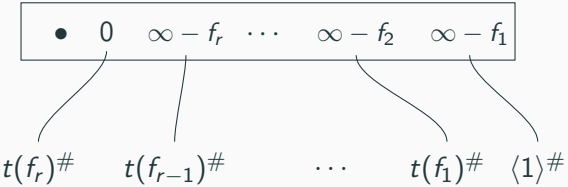
Ideal n-tree I:



Dual ideal $I^{\#}$:



Perspectives for future works

- Efficient method to find minimal generators (benchmarks)
- Find the index of the divisor.
- Use of MDDs in Weyl algebras? (non-commutative spaces)